

Machine Learning

Homework 2

Nov. 2025

Fill your Name
Fill your Student ID

1. For the support vector machine (SVM), the data is not nearly separable in the real world application, so we need to modify SVM by introducing an error margin that must be minimized.

Specifically, the formulation we have looked at is known as the ℓ_1 norm soft margin SVM. In this problem we will consider an alternative method, known as the ℓ_2 norm soft margin SVM. This new algorithm is given by the following optimization problem (notice that the slack penalties are now squared):

$$\begin{aligned} \min_{w,b,\xi} \quad & \frac{1}{2} \|w\|^2 + \frac{C}{2} \sum_{i=1}^N \xi_i^2 \\ \text{s.t.} \quad & y^{(i)}(w^T x^{(i)+b}) \geq 1 - \xi_i, i = 1, \dots, N \end{aligned} \tag{1}$$

- (a) Notice that we have dropped the $\xi_i \geq 0$ constraint in the ℓ_2 problem. Show that these non-negativity constraints can be removed. That is, show that the optimal value of the objective will be the same whether or not these constraints are present.
- (b) What is the Lagrangian of the ℓ_2 soft margin SVM optimization problem?
- (c) Minimize the Lagrangian with respect to w , b , and ξ by taking the following gradients: $\nabla_w \mathcal{L}$, $\frac{\partial \mathcal{L}}{\partial b}$ and $\nabla_\xi \mathcal{L}$, and then setting them equal to 0. Here, $\xi = [\xi_1, \dots, \xi_N]^T$.
- (d) What is the dual of the ℓ_2 soft margin SVM optimization problem?

Ans:

- (a) Consider a potential solution to the above problem with some $\xi < 0$. Then the constraint $y^{(i)}(w^T x^{(i)+b}) \geq 1 - \xi_i$ would also be satisfied for $\xi_i = 0$, and the objective function would be lower, proving that this could not be an optimal solution.

(b)

$$\mathcal{L}(w, b, \xi, \alpha) = \frac{1}{2}w^T w + \frac{C}{2} \sum_{i=1}^N \xi_i^2 - \sum_{i=1}^N \alpha_i [y^{(i)}(w^T x^{(i)} + b) - 1 + \xi_i], \quad (2)$$

where $\alpha_i \geq 0$ for $i = 1, \dots, N$.

(c) Taking the gradient with respect to w , we get

$$0 = \nabla_w \mathcal{L} = w - \sum_{i=1}^N \alpha_i y^{(i)} x^{(i)}, \quad (3)$$

which gives us $w = \sum_{i=1}^N \alpha_i y^{(i)} x^{(i)}$. Taking the derivative with respect to b , we get

$$0 = \frac{\partial \mathcal{L}}{\partial b} = - \sum_{i=1}^N \alpha_i y^{(i)}, \quad (4)$$

giving us $0 = \sum_{i=1}^N \alpha_i y^{(i)}$. Finally, taking the gradient with respect to ξ , we have

$$0 = \nabla_\xi \mathcal{L} = C\xi - \alpha, \quad (5)$$

where $\alpha = [\alpha_1, \dots, \alpha_N]^T$. Thus, for each $i = 1, \dots, N$, we get

$$C\xi_i = \alpha_i \quad (6)$$

(d) The objective function for the dual is

$$\begin{aligned} W(\alpha) &= \min_{w, b, \xi} \mathcal{L}(w, b, \xi) \\ &= \sum_{i=1}^N \alpha_i - \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N \alpha_i \alpha_j y^{(i)} y^{(j)} (x^{(i)})^T x^{(j)} - \frac{1}{2} \sum_{i=1}^N \frac{\alpha_i^2}{C}. \end{aligned} \quad (7)$$

Then the dual formulation for our problem is

$$\begin{aligned} \max_{\alpha} \quad & \sum_{i=1}^N \alpha_i - \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N \alpha_i \alpha_j y^{(i)} y^{(j)} (x^{(i)})^T x^{(j)} - \frac{1}{2} \sum_{i=1}^N \frac{\alpha_i^2}{C} \\ \text{s.t.} \quad & \alpha \geq 0, i = 1, \dots, N \\ & \sum_{i=1}^N \alpha_i y^{(i)} = 0 \end{aligned} \quad (8)$$

2. Consider the following convex optimization problem:

$$\begin{aligned} \min_{x_1, \dots, x_n} \quad & -\sum_{i=1}^n \log(\alpha_i + x_i) + \sum_{i=1}^n \beta_i x_i \\ \text{s.t.} \quad & x_i \geq 0, \sum_{i=1}^n x_i = 1, \end{aligned} \tag{9}$$

where $\alpha_i > 0$ and β is a parameter.

- (a) Write the KKT conditions for the problem.
- (b) Find the optimal solution of the problem.

Ans:

- (a) Construct the Lagrangian function:

$$L(\mathbf{x}, \boldsymbol{\lambda}, \mu) = -\sum_{i=1}^n \log(\alpha_i + x_i) + \sum_{i=1}^n \beta_i x_i - \sum_{i=1}^n \lambda_i (x_i) + \mu \left(\sum_{i=1}^n x_i - 1 \right), \tag{10}$$

where $\mathbf{x} = [x_1, \dots, x_n]^\top$, $\boldsymbol{\lambda} = [\lambda_1, \dots, \lambda_n]^\top \in \mathbb{R}^n$, and $\mu \in \mathbb{R}$.

Therefore, the KKT conditions are:

1. $\forall i, x_i^* \geq 0$ and $\sum_{i=1}^n x_i^* = 1$
2. $\forall i, \lambda_i^* \geq 0$
3. $\forall i, \lambda_i^* x_i^* = 0$
4. $\forall i, \nabla_{x_i^*} L(\mathbf{x}^*, \boldsymbol{\lambda}^*, \mu^*) = \beta_i - \frac{1}{\alpha_i + x_i^*} - \lambda_i^* + \mu^* = 0$

- (b) According to the KKT conditions, we have

$$x_i^* \left(\mu^* + \beta_i - \frac{1}{\alpha_i + x_i^*} \right) = 0, i = 1, \dots, n \tag{11}$$

$$\mu^* \geq \frac{1}{\alpha_i + x_i^*}, i = 1, \dots, n \tag{12}$$

1. If $\mu^* < \frac{1}{\alpha_i} - \beta_i$ and $\beta_i + \mu^* > 0$, Eq.(12) is satisfied when $x_i^* > 0$. Then, according to Eq.(11), we have $x_i^* = \frac{1}{\mu^* + \beta_i} - \alpha_i$;
2. If $\mu^* \geq \frac{1}{\alpha_i} - \beta_i$, according to Eq.(11), we have $x_i^* = 0$.

Thus we have,

$$x_i^* = \max\{0, \frac{1}{\mu^* + \beta_i} - \alpha_i\}. \quad (13)$$

Substituting Eq.(13) into the condition (2), we can obtain

$$\sum_{i=1}^n \max\{0, \frac{1}{\mu^* + \beta_i} - \alpha_i\} = 1. \quad (14)$$

Therefore, the optimal solution of the problem is $x_i^* = \max\{0, \frac{1}{\mu^* + \beta_i} - \alpha_i\}$, $i = 1, \dots, n$, where μ^* satisfies Eq.(14)

3. Given four binary classification datasets, SVMs with different kernel functions were used for classification, and the following decision boundaries were obtained. Please analyze the characteristics of each decision boundary, determine the type of kernel function used, and explain your reasoning.
 - (a) A straight line perfectly separates two types of data.
 - (b) Smooth curved boundaries, presenting as circles or ellipses.
 - (c) The polygonal boundary has distinct corners.

Ans:

- (a) Linear kernel function $K(x_i, x_j) = x_i^T x_j$. Since this data is linearly separable, the linear kernel is sufficient to construct the optimal hyperplane, and the computational complexity is the lowest. Moreover, the support vectors are located on the two separation boundaries, and the number is relatively small.
- (b) Gaussian kernel (RBF kernel) $K(x_i, x_j) = \exp(-\frac{\|x_i - x_j\|^2}{2\sigma^2})$. Because the Gaussian kernel can generate smooth nonlinear boundaries, it is suitable for handling complex nonlinear separable data. The decision boundary depends on σ , which controls the width of the kernel function: When σ is large, the decision boundary is smoother and approximates a linear one. Conversely, when σ is small, the decision boundary is more complex and may lead to overfitting.
- (c) Polynomial kernel $K(x_i, x_j) = (x_i^T x_j + c)^d$. Since the polynomial kernel can generate polynomial decision boundaries, and the roots of the polynomial correspond to the corners of the boundary. Moreover, the decision boundary depends on the influence of d :

- When $d = 1$, the polynomial kernel degenerates into the linear kernel.
- When $d = 2$, the quadratic curve boundary.
- As d increases, the boundary becomes more complex, the fitting ability improves, but it may also overfit.